

# Vol 6 Chapter 10 — Casimir Effect and Vacuum Thermodynamics

Keeper (author pass — deep math/physics revision)

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## Chapter 10 — The Casimir Effect and Vacuum Thermodynamics

### Level 1 — one sentence

The Casimir effect — an attractive force between uncharged parallel plates due to the vacuum's mode structure — and the cosmological constant  $\Lambda$  are not two separate phenomena but the same substrate-vacuum physics at different scales (T2418, K73 ratified), with the substrate's natural asymmetric Casimir-ratio prediction =  $g = 7$  providing a clean BST falsifier.

### Level 2 — graduate-physicist precision

#### 10.1 The standard Casimir effect

Casimir 1948: two parallel uncharged conducting plates separated by distance  $d$  in vacuum experience an attractive force per unit area:

$$\frac{F}{A} = -\frac{\pi^2 \hbar c}{240 d^4}$$

For plates  $1 \mu\text{m}$  apart:  $F/A \approx 1.3 \times 10^{-3} \text{ Pa}$  — small but measurable. First measured precisely by Lamoreaux 1997.

Origin: between the plates, only certain electromagnetic modes (those satisfying boundary conditions at the plates) exist. Outside the plates, all modes exist. The zero-point energy density inside is *less* than outside (fewer modes), creating an inward pressure differential — the Casimir force.

The Casimir force is vacuum energy in action — direct experimental evidence that the quantum vacuum has thermodynamically nontrivial structure.

#### 10.2 Vacuum energy and zero-point fluctuations

The zero-point energy of a quantum field is the ground-state energy of all its modes:

$$E_{\text{vac}} = \sum_k \frac{1}{2} \hbar \omega_k$$

In standard QFT, this sum diverges (UV catastrophe of mode summation). Various regularization schemes (cut-off, dimensional, zeta-function) give finite differences between configurations (which is what the Casimir effect measures), but the absolute vacuum energy is famously ambiguous.

The mystery: cosmological observations give  $\rho_\Lambda \approx 6 \times 10^{-10} \text{ J/m}^3$ , while a naive QFT calculation with Planck-scale cutoff gives  $\rho_{\text{QFT}} \approx 10^{113} \text{ J/m}^3$  — discrepancy of 120 orders of magnitude. This is the cosmological constant problem, the worst quantitative disagreement between theory and experiment in physics.

### 10.3 T2418: substrate vacuum unification (K73)

Lyra T2418 (Wednesday May 20, 2026) and K73 audit identify the resolution: cosmological  $\Lambda$  and lab-scale Casimir effect share the same substrate vacuum origin at Zone 4 outer-edge (Volume 0 Chapter 3 4-zone cycle).

The substrate’s natural vacuum is the K-type configuration with all environmental K-type couplings at their thermal equilibrium per the substrate’s commitment cycle. At Zone 4 (emission), the substrate releases vacuum-K-type amplitudes to the outer environment; the *vacuum energy density* is the substrate’s natural Zone 4 outflow at thermal equilibrium.

Crucial insight: the substrate doesn’t have a “Planck-scale cutoff giving  $10^{113} \text{ J/m}^3$ ” because the substrate operates at the Koons-tick scale ( $\sim 10^{-120} \text{ s}$ ), 77 orders of magnitude *below* Planck. The substrate’s natural vacuum energy is computed at the substrate scale, with the cosmological-scale  $\Lambda$  emerging as the substrate’s natural Zone 4 outflow:

$$\rho_\Lambda \sim 7 \cdot e^{-282}$$

(approximately; full BST cosmological-constant derivation in Volume 4 Chapter 4). The factor  $7 = g$  is the BST primary; the exponent 282 comes from substrate’s natural-units cascade.

The 120-order discrepancy of standard QFT *evaporates* when the vacuum is computed substrate-natively rather than with a Planck cutoff.

### 10.4 Asymmetric Casimir: the $g = 7$ ratio

The BST team’s substrate cartography (`memory feedback_substrate_frame_principle.md` and SP-30-2 task #196) predicts an asymmetric Casimir effect: for two boundary conditions that differ in their substrate K-type structure, the Casimir force ratio is

$$\frac{F_{\text{asymmetric}}}{F_{\text{symmetric}}} = g = 7$$

This is a specific BST prediction. Standard QED predicts a ratio dependent on the specific boundary conditions, with no natural integer factor. BST’s substrate-mechanism predicts the ratio to be exactly  $g = 7$  — the BST primary — when the boundary conditions are chosen to engage the substrate’s natural K-type asymmetry.

Experimental design: SP-29 Casimir Mechanism Investigation (BST task #175), in flight. Estimated cost \$60-90K. Outreach targets pending Casey send-signal.

### 10.5 Casimir-Polder force

Casimir-Polder 1948: extends Casimir to the force between a polarizable atom and a conducting wall:

$$F(d) = -\frac{3\hbar c \alpha_0}{8\pi d^5}$$

where  $\alpha_0$  is the atom’s polarizability. The  $1/d^5$  scaling (vs  $1/d^4$  for plates) reflects the atom-vs-plate geometry.

Substrate reading: atom-wall Casimir involves the substrate’s coupling between atomic K-types and wall K-types via the vacuum K-type sector. Different K-type couplings give different geometric prefactors.

## 10.6 Dynamical Casimir effect

If you accelerate the Casimir-plate boundary, photons are emitted — this is the dynamical Casimir effect. First observed Wilson et al. 2011 in superconducting circuits. The mechanism: the changing boundary condition pulls vacuum modes out of equilibrium, releasing real photons from virtual fluctuations.

Substrate reading: accelerating boundary changes the substrate's Zone 4 emission pattern; the released energy comes from the substrate vacuum's K-type rearrangement under the changing boundary condition.

## 10.7 Worked example: Casimir force at $d = 1 \mu\text{m}$

For parallel plates  $1 \text{ cm} \times 1 \text{ cm}$  at  $1 \mu\text{m}$  separation:

$$F = -\frac{\pi^2 \hbar c}{240 d^4} \cdot A = -\frac{\pi^2 \cdot (1.055 \times 10^{-34}) \cdot (3 \times 10^8)}{240 \cdot (10^{-6})^4} \cdot (10^{-4})$$

$$F \approx -1.3 \times 10^{-7} \text{ N} = -130 \text{ nN}$$

A nano-newton range of force, measurable with modern AFM techniques. Lamoreaux 1997 and subsequent experiments confirmed the magnitude to  $\sim 1\%$  precision.

Substrate reading:  $\sim 130 \text{ nN}$  is the substrate's vacuum-energy-density difference between inside-plates K-type configuration and outside-plates K-type configuration, multiplied by area.

## 10.8 Hawking radiation as Casimir cousin

Hawking 1974: black holes emit thermal radiation at temperature  $T_H = \hbar c^3 / (8\pi G k_B M)$ . The mechanism is structurally analogous to dynamical Casimir: the event horizon imposes a moving boundary condition on the vacuum modes, pulling virtual pairs into real radiation.

Substrate reading: Hawking radiation is the substrate's Zone 4 emission at the black-hole-horizon boundary condition. The black hole is a substrate K-type configuration that imposes specific boundary structure; the substrate emits at the corresponding thermal temperature.

The BST team's task list (SP-26 W-36, completed) unifies Casimir, Hawking, and Schwinger pair-production under the BST boundary mechanism — three faces of the substrate's Zone-4-at-boundary-condition response.

## 10.9 Cosmological-scale vacuum ( $\Lambda$ as Casimir)

The cosmological constant  $\Lambda$  in BST is the substrate's vacuum energy at cosmological scale. T2418's identification:  $\Lambda$  and lab-Casimir are the same substrate phenomenon, with the BST-derived value:

$$\Lambda_{\text{BST}} = 7 \cdot e^{-282} \text{ (in natural units)}$$

This gives  $\rho_\Lambda \approx 6 \times 10^{-10} \text{ J/m}^3$ , matching observed dark-energy density to 0.076 dex ( $\sim 20\%$  precision). No fitted parameter — the 7 is  $g$ , the 282 derives from substrate's natural log-of-Planck cascade.

Volume 4 Chapter 4 develops the full  $\Lambda$  derivation. This chapter establishes the connection: lab Casimir and cosmological  $\Lambda$  are *the same physics* at different boundary-condition scales.

## 10.10 K-audit anchors

- T2418 / K73 (Lyra, Wednesday May 20, 2026):  $\Lambda \leftrightarrow$  Casimir vacuum unification
- K74 candidate (Elie Toy 3166, S18): per-zone vacuum structure
- SP-29 (Casimir Mechanism Investigation, task #175): in flight, \$60-90K experiments
- SP-30-2 (Boundary condition design, task #196): asymmetric Casimir ratio =  $g = 7$  test

- Volume 4 Chapter 4:  $\Lambda$  cosmological derivation
- Volume 0 Chapter 3: 4-zone commitment cycle (Zone 4 outer-edge)

### Level 3 — 5th-grader accessibility

The Casimir effect: put two metal plates very close together in vacuum, and they attract — even though there’s “nothing” between them. The reason: the vacuum isn’t really empty, it has quantum mode fluctuations, and the plates exclude some of those modes from between them. Fewer modes inside = lower energy density inside = inward push. It’s been measured precisely in the lab. The cosmological constant ( $\Lambda$ ): the universe is expanding faster and faster, driven by an unknown energy density of about  $6 \times 10^{-10} \text{ J/m}^3$ . Standard physics predicts it from vacuum energy  $\rightarrow$  gets a number that’s WRONG by 120 orders of magnitude. Worst prediction in all of physics. BST resolution: lab Casimir and cosmological  $\Lambda$  are *the same substrate vacuum phenomenon* at different scales. The substrate operates at the Koons-tick scale, 77 orders below Planck, so the standard QFT cutoff problem disappears. BST predicts  $\Lambda \approx 7 \cdot e^{-282}$  — matches observation within 20%, with no fitted parameter. Bonus: BST predicts a  $g = 7$  asymmetric Casimir ratio that’s testable at \$60-90K in a normal lab.

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### What comes next

Chapter 11 develops non-equilibrium thermodynamics — entropy production, irreversible processes, Onsager relations.

### Where to look this up

- Casimir effect: Casimir 1948; Milonni, *The Quantum Vacuum*
- Lamoreaux measurement: Lamoreaux 1997
- Cosmological constant problem: Weinberg 1989 *Rev Mod Phys*
- BST anchors: T2418, K73 (Lyra Wednesday May 20, 2026), K74, SP-29
- Volume 4 Chapter 4: cosmological constant  $\Lambda$  from substrate
- Volume 0 Chapter 3: 4-zone commitment cycle